

Quantifying the Effect of pH on Tyrosinase Activity Using a Spectrophotometric Assay

Michael Torbensen (4th year BS Environmental Science major)
University of California Santa Barbara, Biology-MCDB1LL

Abstract

Tyrosinase is a copper-containing enzyme central to the melanin pathway of fungi, where it plays roles in pigmentation and a range of defense mechanisms. Because enzyme structure and activity depend on the protonation state of amino acids, extreme acidic or alkaline environments can alter active site chemistry and reduce catalytic efficiency. The hypothesis stated that if a pH of 4-7 is optimal for tyrosinase activity, then any pH below or above will denature the enzyme or stop activity. To test this, a spectrophotometric assay was conducted in which oxidation of L-DOPA → dopaquinone was monitored at 475 nm across three treatments (pH 4, pH 7, and pH 9). The results revealed a quantitative analysis of tyrosinase activity with statistical tests confirming significant differences among treatments. For pH 4, results showed negligible activity, −0.000205 ΔA475/min*μL. For pH 7, results showed moderate activity, 0.000698 ΔA475/min*μL. For pH 9, the highest activity was recorded at 0.002042 ΔA475/min*μL. A two-tailed, two-sample t-test comparing pH 7 to pH 4 obtained a p-value of 0.0014 while another test comparing pH 7 to pH 9 obtained a p-value of 0.000036. Since both of the values were less than 0.05, the findings were statistically significant. These findings supported the hypothesis that tyrosinase activity is strongly inhibited by acidic conditions and enhanced in neutral to alkaline environments. The observed pattern likely reflects changes in the protonation state of amino acid residues and their influence on the active site and metal ion incorporation. Overall, this study acutely demonstrated that tyrosinase activity is pH dependent, maximizing under alkaline conditions while being dramatically reduced in acidic conditions. These findings suggest broader implications for fungal adaptation, since environmental pH shifts can disrupt enzyme regulation and key metabolic pathways required for survival. Understanding these dynamics not only highlights mechanisms of fungal stress tolerance and homeostasis but also provides insights with potential applications in and beyond biotechnology.

Introduction

Tyrosinase is a crucial enzyme that catalyzes the oxidation of tyrosine in the melanin pathway of fungi, producing pigmentation for defense, wound repair, and UV protection. Its enzymatic activity can be influenced by a variety of environmental conditions. Optimal temperatures allow tyrosinase to catalyze quickly and anything outside of its optimal range will slow down or stop activity. Availability of copper ions are a necessity to carry out redox reactions and continue activity. Because fungal cells are also sensitive to extracellular pH, they have mechanisms that maintain homeostasis. The mechanisms include proton pumps such as Pma1 and V-ATPases to maintain conditions that support enzyme function (Chen Y-P, 2025). The specific range of optimal pH for maximum tyrosinase activity is not known. This study investigated the idea of how the enzymatic activity of tyrosinase would be affected if put in various solutions of varying pH. These findings helped to support the many uses for tyrosinases including melanoma treatment, as biosensors, and synthesis of medical hydrogels (Pretzler M. 2024). The results also provided evidence for how tyrosinase activity can be affected in the natural environment. A series of tests were conducted in order to quantify tyrosinase activity changes in low (pH 4), neutral (pH 7), and high (pH 9) conditions. Spectrophotometry was used to identify and quantify the activity of tyrosinase. The hypothesis stated that if a pH of 4-7 is optimal for tyrosinase activity, then any pH below or above will denature the enzyme or stop activity.

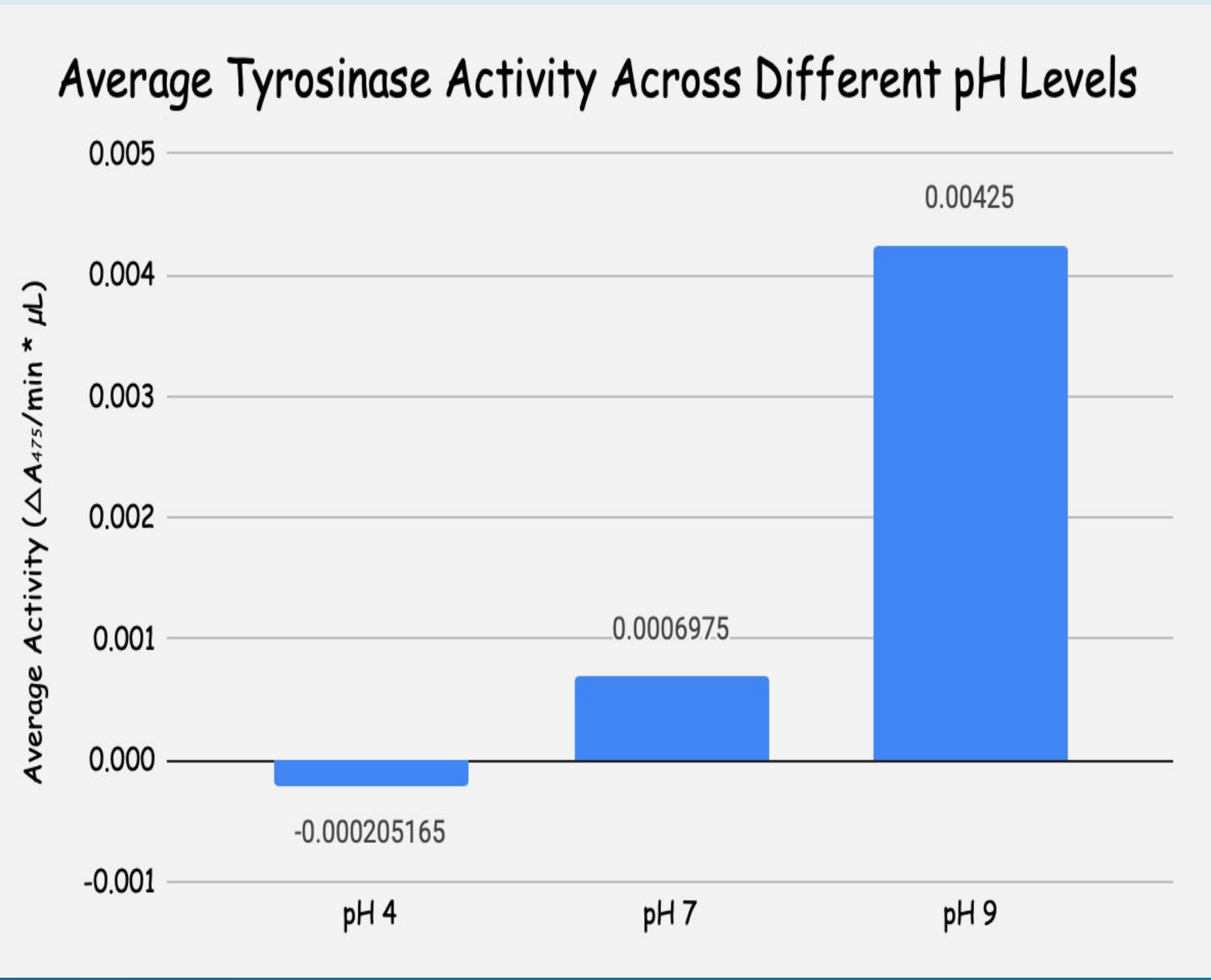
References & Acknowledgements

I would like to thank my professor Kris Troy for, not only inspiring me with my love for biology, but helping me understand and master the basic concepts of biology. This has had very positive effects on my ability to smoothly convey scientific concepts to a broader audience. I would also like to thank my TA, Amanda Maheras, for her enthusiastic energy in our laboratory. It would not have been possible to complete this project without the emotional support of my daily gardening habits too!

- Nguyen A, MCDB 1LL University of California Santa Barbara, 2025. <https://bookshelf.vitalsource.com/reader/books/9781533973009>
- Bin B-H, Bhin J, Yang SH, Shin M, Nam Y-J, Choi D-H, Shin DW, Lee A-Y et al. 2015. Membrane-associated transporter protein (matp) regulates melanosomal ph and influences tyrosinase activity. doi:10.1371/journal.pone.0129273
- Chen Y-P, Li M, Liu Z, Wu J, Chen F, Zhang S. 2025. Inhibition of tyrosinase and melanogenesis by carboxylic acids: Mechanistic insights and safety evaluation. Molecules (Basel, Switzerland). 30(7):1642.
- Kane PM. 2016. Proton transport and ph control in fungi. Advances in experimental medicine and biology. doi:10.1007/978-3-319-25304-6_3
- Maria-Solano MA, Ortiz-Ruiz CV, Muñoz-Muñoz JL, Teruel-Puche JA, Berna J, Garcia-Ruiz PA, Garcia-Canovas FhooX, Garcia-Canovas F. 2016. Further insight into the ph effect on the catalysis of mushroom tyrosinase. Journal of molecular catalysis. doi:10.1016/j.molcatb.2015.12.008
- Pretzler M, Rempel A. 2024. Tyrosinases: A family of copper-containing metalloenzymes. Chemtexts. doi:10.1007/s40828-024-00195-y

Results

Average tyrosinase activity ($\Delta A_{475}/\text{min} \cdot \mu\text{L}$) was quantified using an enzymatic assay in which oxidation of L-DOPA to dopachrome was observed spectrophotometrically at 475 nm under three pH treatments. At first, levels of tyrosinase activity was measured in a pH 7 solution containing 2.6mL 100 mM KPO_4 + 0.35mM SDS, 0.4mL L-DOPA and 200 μL mushroom homogenate (Nguyen A. 2024). After triplicate spectrophotometry trials, the average enzymatic activity at was measured at $0.0006975 \Delta A_{475}/\text{min} \cdot \mu\text{L}$. This value was positive, indicating that tyrosinase was able to catalyze the oxidation of L-DOPA under neutral conditions, but the amount of activity remained relatively low. Replicates consistently produced linear increases in absorbance with moderate R^2 values, confirming that the activity was not just random fluctuations. On the contrary, enzymatic activity was then measured in a pH 4 solution containing 2.6mL 100mM citrate buffer + 0.35mM SDS, 0.4mL L-DOPA and 200 μL mushroom homogenate (Nguyen A. 2024). After another round of triplicate spectrophotometry trials, average tyrosinase activity was measured at $-0.000205165 \Delta A_{475}/\text{min} \cdot \mu\text{L}$. Replicates consistently produced flat or negative slopes with very low R^2 values, confirming that the enzyme was unable to catalyze oxidation under acidic conditions. Statistical analysis using a two-tailed, two-sample t-test showed that the difference between pH 7 and pH 4 was highly significant ($p = 0.0014$) and supported the conclusion that activity at neutral pH is notably greater than at acidic pH. Lasty, enzymatic activity was measured in a pH 9 solution containing 2.6mL Tris-HCL buffer + 0.35mM SDS, 0.4mL L-DOPA and 200 μL mushroom homogenate (Nguyen A. 2024). These alkaline conditions provided the greatest amount of tyrosinase activity observed, where average values ($0.002042 \Delta A_{475}/\text{min} \cdot \mu\text{L}$) were nearly three times greater than what was observed at pH 7. Replicates consistently showed strong linear relationships between time and absorbance, producing high R^2 values. A two-tailed, two-sample t-test confirmed that the difference between pH 9 and pH 7 was also highly significant ($p = 0.000036$), and much more statistically significant than the difference between pH 7 and pH 4. Together, these findings quantitatively demonstrated that tyrosinase activity increases with pH, as shown in Figure 1, and supported the conclusion that alkaline conditions strongly enhance enzyme function compared to neutral or acidic environments.



Discussion & Conclusion

A recent study stated that tyrosinase activity peaks around pH 6–7 and anything outside that range begins to decline (Chen Y-P, 2025). Although, the results of this study demonstrated that tyrosinase activity is highest at alkaline pH (9), moderate at neutral pH (7), and negligible at acidic pH (4). These findings supported the hypothesis that tyrosinase functions optimally in alkaline conditions, while its activity is severely inhibited in acidic environments. Although the hypothesis was supported, there may have been potential sources of error. Variations in collected mushroom homogenate solutions could have caused slight differences in total enzyme concentration between each trials. In addition, minor changes in buffer pH during the assay due to duration of tests or spectrophotometric inconsistencies may have contributed to noise in the data. Although these factors didn’t alter the overall trend, they are still possible sources of slight error. Since statistically significant differences ($p < 0.05$) were observed between all tested pH levels, the results concluded that if tyrosinase is exposed to acidic conditions its activity decreases, whereas if it is exposed to alkaline conditions its activity increases, demonstrating that pH strongly regulates enzyme function. A 2016 study showed that the charge status of histidine in the active site of tyrosinase could be altered by acidic conditions. This could hinder the incorporation of the copper metal ion into the active site by possibly deforming it and subsequently decreasing tyrosinase activity (Kane P.M. 2016). On the other hand, at physiological pH, glutamic acid is deprotonated and able to interact with phenolic hydroxyl groups of the substrate attracting and orienting it *towards* the active site of tyrosinase, facilitating access and activity (Kane P.M. 2016). So, when pH is acidic, glutamic acid is in a protonated state, and tyrosinase activity is also severely decreased (Maria-Solano MA. 2016). The experimental results were closely in line with this statement as one can see that tyrosinase activity is dramatically increased when exposed to alkaline pH. The presented results also aligned with previous studies showing that, since tyrosinase is sensitive to extracellular pH, it is essential that proton gradients need to be maintained by the proton pump Pma1 and V-ATPases (Chen Y-P). They actively pump protons out of the cell or into vacuoles to counteract external pH changes. Without the presence of these active homeostatic regulators, extracellular pH changes can have a series of effects on tyrosinase activity and the overall health of the organism. This knowledge, including the results at hand, could inform strategies to manipulate fungal melanization—for example, in biocontrol (inhibiting melanin to weaken pathogens), biomedical applications, or even the food industry. The next logical step would be to examine tyrosinase activity in a much more narrow pH range to better understand the optimum. Further studies could even measure the combined effect of temperature and pH on tyrosinase activity.

In conclusion, the results contradicted the initial assumption that the enzyme would function best between pH 4 and 7, although it did support the prediction that acidic conditions would strongly inhibit tyrosinase activity. Instead, elevated activity was observed at pH 9, which indicated that alkaline conditions enhance enzyme function more effectively than hypothesized. Thus, the hypothesis should be re-evaluated to reflect that tyrosinase activity is optimized under alkaline rather than neutral conditions.